

NANYANG TECHNOLOGICAL UNIVERSITY**SEMESTER 1 EXAMINATION 2023-2024****IE4476 – IMAGE PROCESSING**

November / December 2023

Time Allowed: 2 hours

INSTRUCTIONS

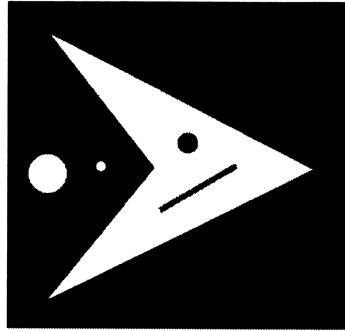
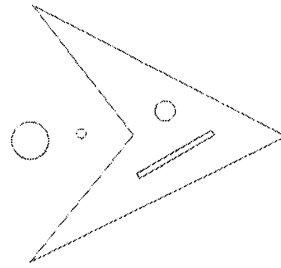
1. This paper contains 4 questions and comprises 3 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a Restricted Open Book Examination. You are allowed to bring into the examination hall one double-sided A4-size reference sheet with texts and figures handwritten, drawn, or printed on the A4 paper (no sticky notes/post-it notes on the reference sheet).
5. Unless specifically stated, all symbols have their usual meanings.

1. An image $f(x, y)$ is degraded by a linear, space-invariant process $h(x, y)$ and noise $\eta(x, y)$ in the spatial domain as

$$g(x, y) = h(x, y) * f(x, y) + \eta(x, y)$$

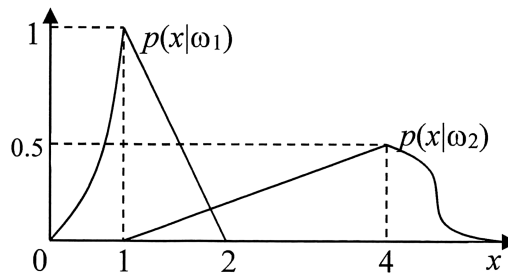
- (a) Express the image degradation model in the frequency domain. (5 Marks)
- (b) Design a generalized inverse filter $R(u, v)$ in the frequency domain for image restoration from the degraded image $g(x, y)$ so that $|R(u, v)| < 10$. (12 Marks)
- (c) Supposing that the power spectra of the image $f(x, y)$ and noise $\eta(x, y)$ are respectively $S_f(u, v) = \frac{1}{u^2 + v^2}$ and $S_\eta(u, v) = 0.5$, design a Wiener filter for the image restoration. (8 Marks)

2. A binary image $f(x, y)$ shown in Figure 1 has value 1 for white pixels and 0 for black pixels. The boundary of the image is plotted in Figure 2. A LSI filter mask $h(x, y)$ has value 1 inside a circle and 0 outside the circle. The area of the filter mask (circle) is 63. The diameter d of the filter mask is larger than that of the smallest circle and the width of the strip in Figure 1 but smaller than those of the two larger circles in Figure 1. Let $g(x, y)$ be the filter response to the input image $f(x, y)$, and $g_{\min}$ and $g_{\max}$ be the minimum and maximum values of $g(x, y)$, respectively.

**Figure 1****Figure 2**

- (a) What are the minimum value $g_{\min}$ and maximum value $g_{\max}$ of $g(x, y)$? (4 Marks)
- (b) Let $g_1(x, y)$ be the binarized image of $g(x, y)$ by setting $g_1(x, y)=1$ if $g(x, y) > g_{\min}$, otherwise $g_1(x, y) = 0$. Copy Figure 2 to your answer script and plot the boundary of $g_1(x, y)$ on it. (7 Marks)
- (c) Let $g_2(x, y)$ be the binarized image of $g(x, y)$ by setting $g_2(x, y)=0$ if $g(x, y) < g_{\max}$, otherwise $g_2(x, y) = 1$. Copy Figure 2 to the answer script and plot the boundary of $g_2(x, y)$ on it. (7 Marks)
- (d) As $f(x, y)$, $h(x, y)$, $g_1(x, y)$ and $g_2(x, y)$ all are binary images, which can be represented by pixel sets, express $g_1(x, y)$ and $g_2(x, y)$ by $f(x, y)$ and $h(x, y)$ in terms of set operation. (7 Marks)
3. An image contains 40 fishes of type ω_1 and 60 fishes of type ω_2 . You need to identify the type of each fish in the image. An image segmentation algorithm is used to separate each fish and compute its size. The two probability density functions (PDFs) of the size x of the two types of fishes, $p(x|\omega_1)$ and $p(x|\omega_2)$, are shown in Figure 3 on page 3, where 2 straight lines can be observed.

Note: Question No. 3 continues on page 3.

**Figure 3**

- (a) Using optimal thresholding to determine the type of a fish with $x = 1.5$, which minimizes the probability of wrong decision, and calculate the error rate. (12 Marks)
- (b) Generate a decision rule to determine the type of fish based on x , which minimizes the probability of wrong decision, and evaluate the error rate of your method. (13 Marks)
4. Multilayer perceptron (MLP) is a fully connected feed-forward neural network, which is the foundation of various deep learning approaches.
- (a) Draw the network architecture of a typical two-layer MLP that has two inputs x_1 and x_2 , two hidden neurons, and one output neuron. Show all network parameters in the diagram. (6 Marks)
- (b) The hidden neurons have the nonlinear activation function $f(s) = \frac{1}{1 + \exp(-s)}$, and the output neuron has the linear activation function $g(s) = s$. Express the output y of the network as a function of the two inputs x_1 and x_2 . (6 Marks)
- (c) Supposing the desired output for the input vector $[x_1 \ x_2]^T$ is t , derive a learning rule that adjusts the parameter matrix V of the second layer to make the network output closer to the desired output in a step-by-step manner. (13 Marks)

END OF PAPER

IE4476 IMAGE PROCESSING & COMPUTER VISION

Please read the following instructions carefully:

- 1. Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.